

10/05/2026: Midterm Review Part 2: More Proofs

CSCI 246: Discrete Structures

October 5, 2026

Announcements

- Let's review some midterm logistics.
- If you prepared for today's optional quiz retake, you're 1/5 of the way done studying for the midterm! Yippee!

Today's Agenda

- Midterm discussion (5 mins)
- Optional Quiz Retake (20 mins)
- Group exercises (25 mins)

Group exercises

Ashley, Isabelle: 7	Hager, Nathan: 11	Putnam, John: 5
Bailey, Tanna: 7	Horner, Jayden: 7	Ramos, Vydel: 10
Blaisdell, Olyver: 5	Johnson, Jaret: 3	Rathbun, Jedidiah: 9
Brown, Ashton: 1	Kintzel, Samantha: 1	Roberts, Matthew: 13
Brown, Jonathan: 2	Lameres, Alexis: 9	Roller, Nathan: 4
Cardenas, Dominic: 2	Lee, Holden: 5	Ryan, Otis: 3
Cutler, George: 11	Mayala, Reece: 6	Shevtsov, Timothy: 1
Fabik, Roland: 8	McLean, Connor: 12	Sinclair, Aidan: 11
Fellows, Audrey: 10	Mcbride, Daniel: 10	Smith, Charles: 6
Fitzgerald, Ryan: 8	Miller, Maverick: 2	Stensland, Emma: 13
Fuller, Ryan: 6	Moss, Emily: 8	Thiessen, Logan: 9
Griffen, Clara: 4	Mousseau, Matthew: 3	Thompson, Jaiden: 4
Grutsch, Tanner: 12	Neuman, Evan: 13	Vath, Ethan: 12

Group exercises

New group exercises on proofs.

1. Prove that the product of two even integers is even.
2. Let x be an integer. Prove that $0 \mid x$ if and only if $x = 0$.
3. Disprove: If p and q are prime, then $p + q$ is composite.
4. Let $A = \{x \in \mathbb{Z} : 4 \mid x\}$ and let $B = \{x \in \mathbb{Z} : 2 \mid x\}$. Show that $A \subseteq B$.
5. Let $A = \{n \in \mathbb{Z} : 3 \mid n\}$ and let $B = \{n \in \mathbb{Z} : 3 \mid (n + 6)\}$. Show that $A = B$.

Practicing some of the older group exercises on proofs.

1. A **tautology** is a Boolean expression that evaluates to TRUE for all possible values of its variables. Use truth tables to show the following is a tautology:

$$(x \implies y) \wedge (y \implies z) \implies (x \implies z)$$

2. Let n be a positive integer. Prove that congruence modulo n is an equivalence relation on the set of integers.
3. The distributive properties for sets are

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C), \quad \text{and} \quad A \cap (B \cup C) = (A \cap B) \cup (A \cap C).$$

Prove the second property.

Solution: Group exercise #1

Problem. Prove that the product of two even integers is even.

Solution. We show that if $z = xy$ where x and y are even, then z is even. Let x and y be even. Then $x = 2a$ where a is an integer and $y = 2b$ where b is an integer. So $z = xy = (2a)(2b) = (4ab) = 2(2ab)$. Hence $z = 2c$ for some integer c (namely $c = 2ab$). Hence z is even.

Reference: Definition

An integer is called **even** if it is divisible by 2. (That is, an integer z is even if there is an integer c such that $z = 2c$.)

Solution: Group exercise #2

Problem. Let x be an integer. Prove that $0 \mid x$ if and only if $x = 0$.

Solution. First we show that if $0 \mid x$ then $x = 0$. Let $0 \mid x$. Then there is an integer c such that $0c = x$. So $x = 0$.

Now we show that if $x = 0$ then $0 \mid x$. Let $x = 0$. Let c be any integer (e.g. $c = 7$). Then $x = 0c$. Hence, there is an integer c such that $0c = x$. Hence $0 \mid x$.

Reference: Definition

Let a, b be integers. We say that a is **divisible** by b , written $b \mid a$, if there is an integer c such that $bc = a$.

Solution: Group exercise #3

Problem. Disprove: If p and q are prime, then $p + q$ is composite.

Solution. We give a counterexample where p and q are prime, but $p + q$ is not composite. Let $p = 2$ and $q = 3$. Then p and q are both prime. ($2 > 1$ and the only positive divisors of 2 are 1 and 2, and $3 > 1$ and the only positive divisors of 3 are 1 and 3.). However, $p + q = 5$, which is prime ($5 > 1$ and the only positive divisors of 5 are 1 and 5), and therefore not composite.

Reference: Definitions

- An integer p is called **prime** if $p > 1$ and the only positive divisors of p are 1 and p .
- A positive integer a is called **composite** if there is an integer b such that $1 < b < a$ and $b \mid a$.

Solution: Group exercise #4

Problem. Let $A = \{x \in \mathbb{Z} : 4 \mid x\}$ and let $B = \{x \in \mathbb{Z} : 2 \mid x\}$. Show that $A \subseteq B$.

Solution. We will show that if $x \in A$, then $x \in B$. Let $x \in A$. Then $4 \mid x$. So there is an integer c such that $x = 4c$. So $x = 2(2c)$. So $x = 2d$ for some integer d (namely $d = 2c$). So $2 \mid x$. So $x \in B$.

Reference: Definition

Let a, b be integers. We say that a is **divisible** by b , written $b \mid a$, if there is an integer c such that $bc = a$. We call b a **divisor** of a .

Solution: Group exercise #5

Problem. Let $A = \{n \in \mathbb{Z} : 3 \mid n\}$ and let $B = \{n \in \mathbb{Z} : 3 \mid (n + 6)\}$. Show that $A = B$.

Solution. First we show that if $x \in A$, then $x \in B$. Let $x \in A$. Then $3 \mid x$. So $x = 3c$ for some integer c . So $x + 6 = 3c + 6$. So $x + 6 = 3(c + 2)$. So $x + 6 = 3d$ for some integer d , namely $d = c + 2$. So $3 \mid (x + 6)$. So $x \in B$.

Now we show that if $x \in B$, then $x \in A$. Let $x \in B$. So $3 \mid (x + 6)$. So $3c = x + 6$ for some integer c . So $3c - 6 = x$. So $3(c - 2) = x$. So $3d = x$ for some integer d (namely $d = c - 2$). So $x \in A$.

Reference: Definition

Let a, b be integers. We say that a is **divisible** by b , written $b \mid a$, if there is an integer c such that $bc = a$. We call b a **divisor** of a .